LFR maze — five-chip gate wiring

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This is the lowest common-chip build of the motor equations. Five 14-pin ICs. No Arduino. IN4 of the L298N is tied to ground so the right wheel never runs backward.

1. Buy these five chips

Qty	Part	What it is	Gates used
1	74HC04	Hex inverter	6 of 6
2	74HC08	Quad 2-input AND	8 of 8
2	74HC32	Quad 2-input OR	7 of 8 (one spare)

Do not buy a third 74HC08. The sixth inverter on the 74HC04 is not spare: it makes T', which replaces an AND gate. Also needed: breadboard, 5 V for logic, 7.4 V for motors, one common ground, L298N, five TCRT5000 modules.

2. Lames of the wires

Every TCRT5000 is active low. The module pin is 0 when it sees a wall or the black line. Call those pins a b c d e. After one inverter they become A B C D E, which are the bits in the truth table (1 = wall or black).

Module pin	Goes low when	Inverter output	Use the raw pin as
a left wall	wall close	A = NOT a	A'
b front wall	wall ahead	B = NOT b	B'
c right wall	wall close	C = NOT c	C'
d left line	on black tape	D = NOT d	D'
e right line	on black tape	E = NOT e	E'

Never invert a module pin twice. If an equation needs A', wire pin a. If it needs A, wire the 74HC04 output.

3. The four motor equations

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LF = B'(D' + E) + B C'(A + D'E)
LB = BC + A'B(D + E') + B'DE'
RF = B'(D + E') + B(C + A'(D + E'))
RB = 0
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Right turn: left wheel forward, right wheel stopped (LF=1, RF=0). Left turn and dead-end U-turn: left wheel back, right wheel forward (LB=1, RF=1).

4. Shared nets that save the extra AND chip

T = D + E'. Then T' is exactly D'E. That uses the last inverter instead of a ninth AND, so two 74HC08 packages are enough.

```
T = D OR e
T' = NOT T (sixth 74HC04 gate)
S = d OR E
aT = a AND T
U = C OR aT (left spin + U-turn)
W = B AND U
V = A OR T'
Bc = B AND c
BcV= Bc AND V
bS = b AND S
bT = b AND T
bD = b AND D
bDe= bD AND e
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LF = bS OR BcV → L298N IN1
LB = W OR bDe → L298N IN2
RF = bT OR W → L298N IN3
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RB = 0 → L298N IN4 to GND

5. Chip pinout (every 14-pin gate IC is the same layout)

Pin 7 = GND. Pin 14 = +5 V. Decouple each chip with 100 nF from pin 14 to pin 7, next to the IC.

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74HC04: 1A=1 1Y=2 2A=3 2Y=4 3A=5 3Y=6 GND=7 4Y=8 4A=9 5Y=10 5A=11 6Y=12 6A=13 VCC=14
74HC08 and 74HC32: 1A=1 1B=2 1Y=3 2A=4 2B=5 2Y=6 GND=7 3Y=8 3A=9 3B=10 4Y=11 4A=12 4B=13 VCC=14
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Output of a gate is Y. Do not short two Y pins together. One output may fan out to several inputs.

6. Wire U1 — 74HC04 (inverters)

Pin	Name	Connect to
1	1A	sensor a (left wall module)
2	1Y = A	U2 pin 12, and any gate that needs A
3	2A	sensor b (front wall module)
4	2Y = B	U4 pins 4 and 9
5	3A	sensor c (right wall module)
6	3Y = C	U2 pin 9
7	GND	common ground
8	4Y = D	U2 pin 1, U5 pin 10
9	4A	sensor d (left line module)
10	5Y = E	U2 pin 5
11	5A	sensor e (right line module)
12	6Y = T'	U2 pin 13
13	6A	U2 pin 3 (net T). Wire this after U2 exists.
14	VCC	+5 V

7. Wire U2 — 74HC32 #1 (first four ORs)

Pin	Name	Connect to
1	1A	U1 pin 8 (D)
2	1B	sensor e
3	1Y = T	U1 pin 13, U4 pin 2, U5 pin 5
4	2A	sensor d
5	2B	U1 pin 10 (E)
6	2Y = S	U5 pin 2
7	GND	common ground
8	3Y = U	U4 pin 5
9	3A	U1 pin 6 (C)
10	3B	U4 pin 3 (aT)
11	4Y = V	U4 pin 13
12	4A	U1 pin 2 (A)
13	4B	U1 pin 12 (T')
14	VCC	+5 V

8. Wire U3 — 74HC32 #2 (motor ORs)

Pin	Name	Connect to

1	1A	U5 pin 3 (bS)
2	1B	U4 pin 11 (BcV)
3	1Y = LF	L298N IN1
4	2A	U5 pin 6 (bT)
5	2B	U4 pin 6 (W)
6	2Y = RF	L298N IN3
7	GND	common ground
8	3Y = LB	L298N IN2
9	3A	U4 pin 6 (W)
10	3B	U5 pin 11 (bDe)
11	4Y spare	leave open
12	4A spare	GND (so the unused gate does not float)
13	4B spare	GND
14	VCC	+5 V

9. Wire U4 — 74HC08 #1 (first four ANDs)

Pin	Name	Connect to
1	1A	sensor a
2	1B	U2 pin 3 (T)
3	1Y = aT	U2 pin 10
4	2A	U1 pin 4 (B)
5	2B	U2 pin 8 (U)
6	2Y = W	U3 pins 5 and 9
7	GND	common ground
8	3Y = Bc	U4 pin 12
9	3A	U1 pin 4 (B)
10	3B	sensor c
11	4Y = BcV	U3 pin 2
12	4A	U4 pin 8 (Bc)
13	4B	U2 pin 11 (V)
14	VCC	+5 V

10. Wire U5 — 74HC08 #2 (last four ANDs)

Pin	Name	Connect to
1	1A	sensor b
2	1B	U2 pin 6 (S)
3	1Y = bS	U3 pin 1
4	2A	sensor b
5	2B	U2 pin 3 (T)
6	2Y = bT	U3 pin 4
7	GND	common ground
8	3Y = bD	U5 pin 12
9	3A	sensor b
10	3B	U1 pin 8 (D)
11	4Y = bDe	U3 pin 10
12	4A	U5 pin 8 (bD)
13	4B	sensor e
14	VCC	+5 V

11. L298N and power

L298N	From
IN1	U3 pin 3 LF
IN2	U3 pin 8 LB
IN3	U3 pin 6 RF
IN4	GND (RB is always 0)
ENA, ENB	Leave jumpers ON
OUT1, OUT2	Left motor

OUT3, OUT4	Right motor
+12 V / motor VCC	7.4 V pack only
+5 V	Do not use the L298N regulator for the 74HC chips
GND	Join pack negative, 5 V negative, every chip pin 7, every sensor GND

Logic 5 V is a USB power bank or a separate 5 V regulator. Sensors VCC to that 5 V. Never parallel a power-bank 5 V with the L298N 5 V pin.

12. Build order

1. Power rails and 100 nF caps. Check 5 V on every pin 14 before inserting ICs.
2. U1 pins 1–11 only (the five sensor inverters). Leave pin 13 unconnected until T exists.
3. U2 pins 1–6 (T and S). Then U1 pin 13 to U2 pin 3.
4. U4 gate 1 (aT), then U2 gates 3 and 4 (U and V).
5. Rest of U4, all of U5, then U3 to the L298N.
6. Tie U3 spare inputs to ground. Tie IN4 to ground.

13. Bench tests (logic 5 V on, motors optional)

What you do	A B C D E	LF LB RF RB	Must see
Open floor, line in the gap	0 0 0 0 0	1 0 1 0	both wheels forward
Corridor, line in the gap	1 0 1 0 0	1 0 1 0	straight
Line under D only	1 0 1 1 0	0 1 1 0	pivot left
Line under E only	1 0 1 0 1	1 0 0 0	left wheel only
Wall ahead, left open	0 1 1 0 0	0 1 1 0	pivot left
Wall ahead, right open	1 1 0 0 0	1 0 0 0	left wheel only
Boxed in (dead end)	1 1 1 x x	0 1 1 0	U-turn spin

A–E in that table are the inverter outputs. On the modules, a dead end is a=b=c=0 (all three wall pins low). If LF and LB are both high, a Y pin is shorted or a chip is in backwards. Stop and fix that before connecting motors.

Checked against all 32 input rows. Gate count: 6 NOT + 8 AND + 7 OR = 21 gates, 5 packages. A sixth chip would only be needed if you refused to use T' as D'E.